

The background of the slide is a photograph of classical architecture. It features a series of tall, fluted columns in the foreground, creating a sense of depth. In the background, a pediment is visible, adorned with a golden sunburst or starburst design. The lighting is warm, with a golden glow filtering through the columns, suggesting a sunset or sunrise scene.

CodeML

ÉQUIALGO: FAIR STUDENT FINANCING

BIAS DIAGNOSIS, CORRECTION, AND A PROPOSED MONITORING PLAN FOR PRODUCTION

VNPHOEVER

ACCURATE IS NOT FAIR



48.4%

granted in Montréal and Capitale-Nationale

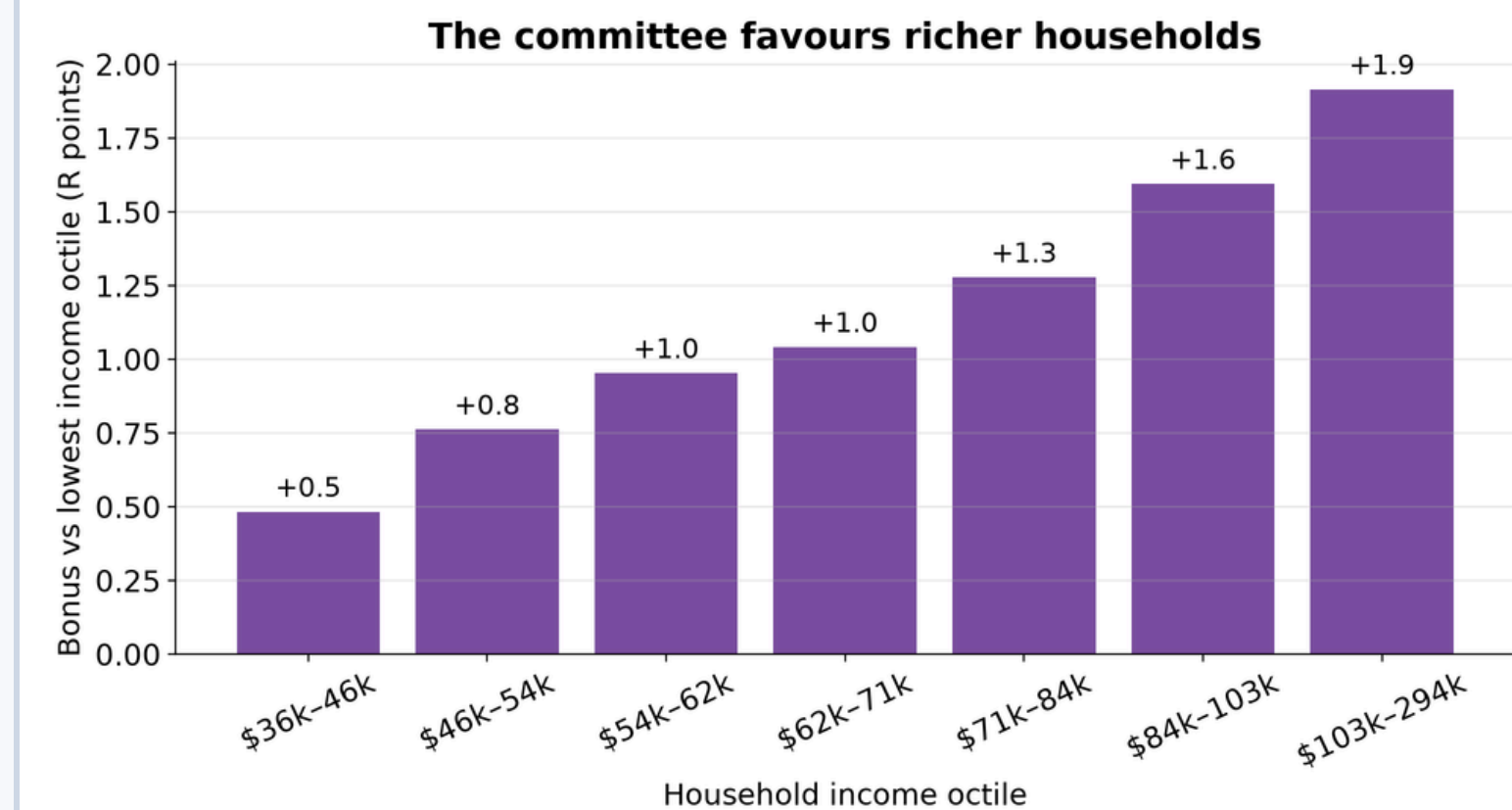
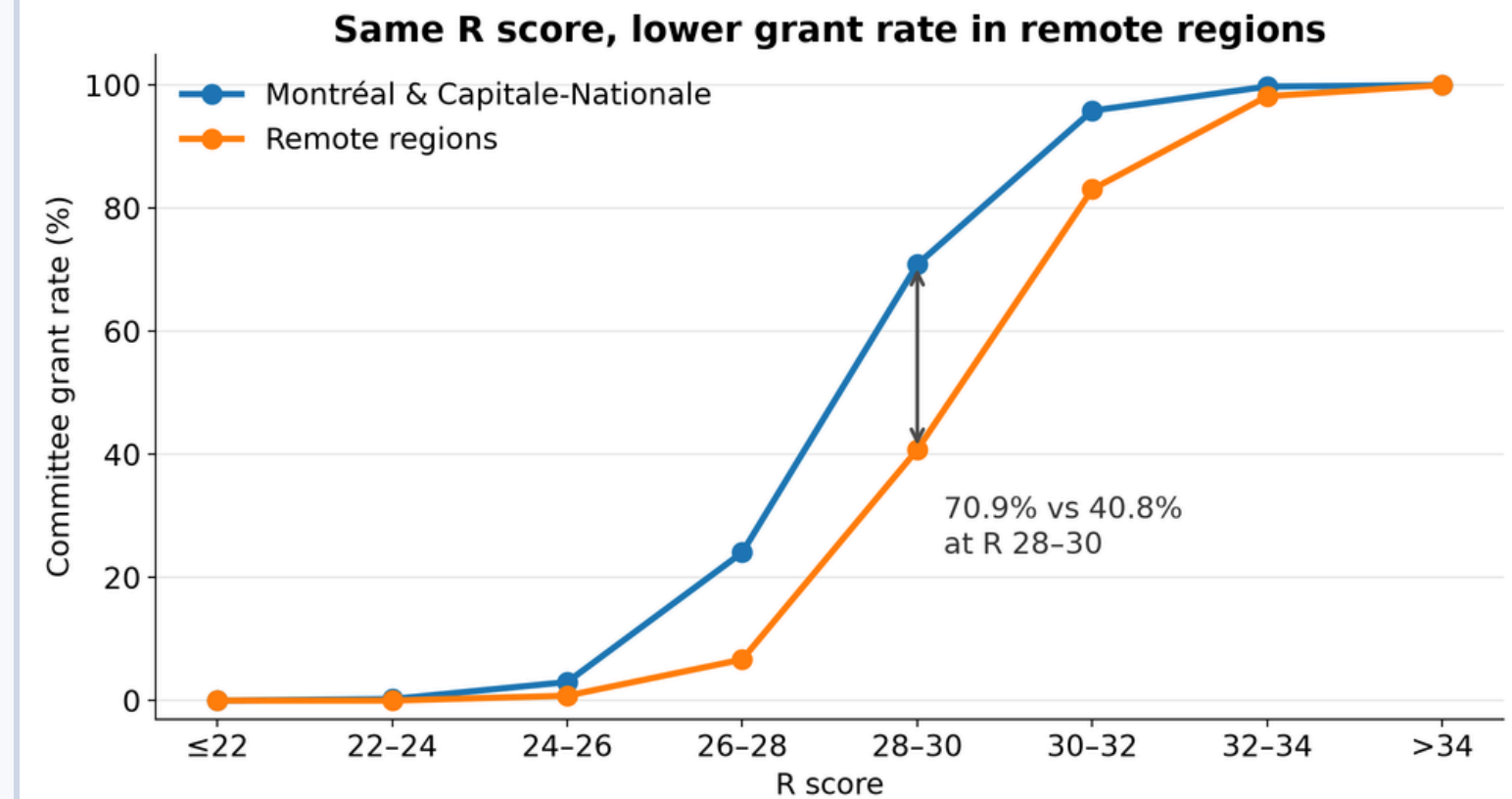
27.3%

granted in the remote regions

The production model is 88% accurate because it faithfully reproduces a committee that grants 48.4% of applicants in Montréal and Capitale-Nationale, and 27.3% in the remote regions.

SAME GRADES, DIFFERENT REGION, DIFFERENT OUTCOME

- At an R score of 28–30: 70.9% granted in the centres vs 40.8% in remote regions
- The committee adds a **1.40 R-point penalty to remote applicants** (95% CI 1.27–1.54)
- A hidden second bias: richer households **get +1.9 R points** (lowest vs highest income octile).



DELETING THE REGION DOESN'T DELETE THE BIAS

Postal code maps 1:1 to the region; distance predicts it with AUC 0.998

- Removing region and postal code:
parity gap 0.188 → 0.173
- Without the region, the penalty **leaks into distance**
(−0.375 R points per 100 km)
- The fairlearn trap: on committee labels, TPRs are
already equal (0.849 vs 0.838). The constraint has
nothing to fix, because the bias is in the labels.

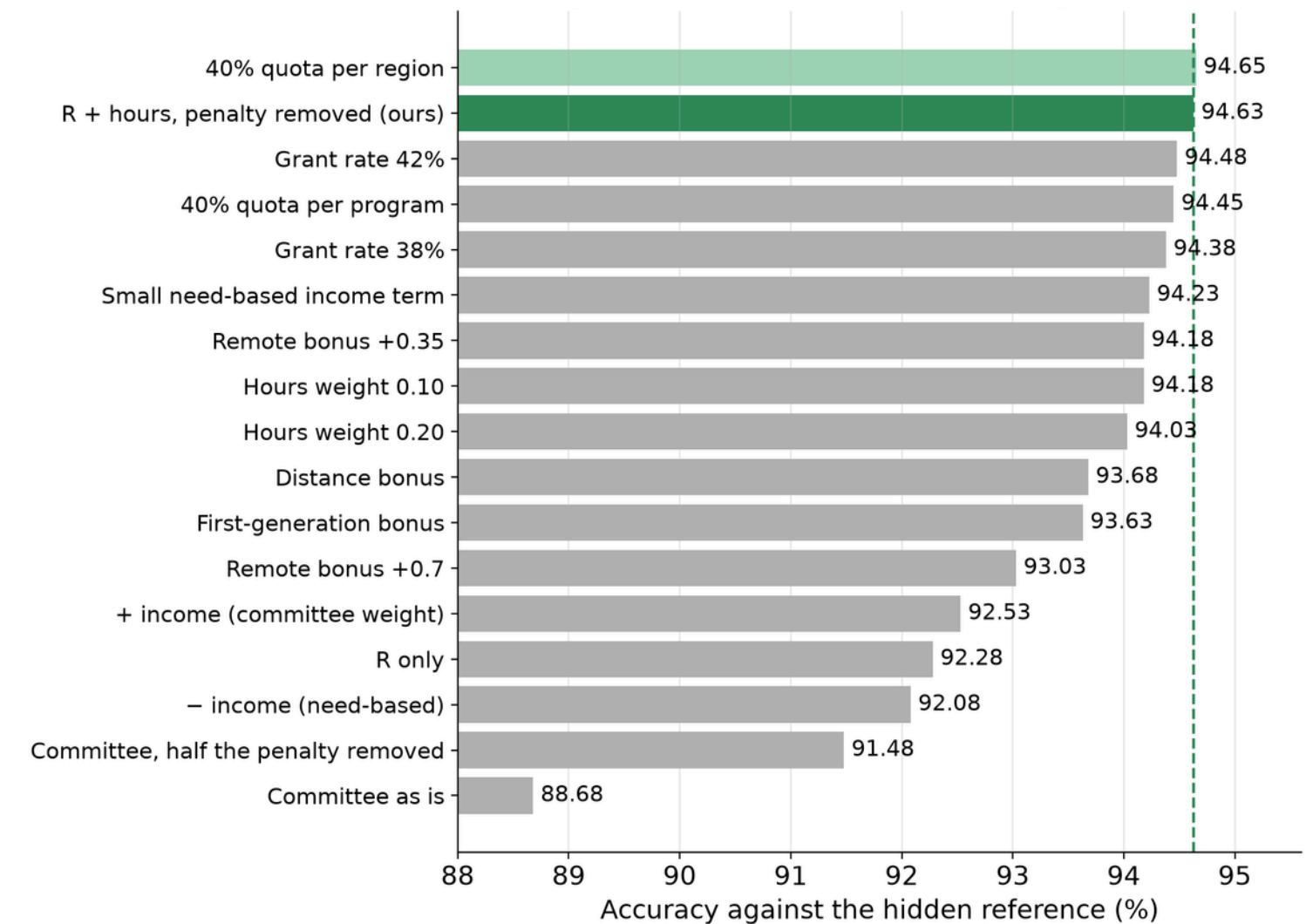
Model inputs	Parity gap
All features	0.188
Without region	0.180
Without region + postal code	0.173

17 EXPERIMENTS TO REVERSE-ENGINEER MERIT

Merit = R score + 0.145 × weekly hours worked

- Each submission changed one element of the rule
- All of these scored worse: income (either direction), distance, remote bonuses, first-generation bonus, quotas
- All 18 probe files are regenerated identically from code

No probe beats "R + hours" beyond noise



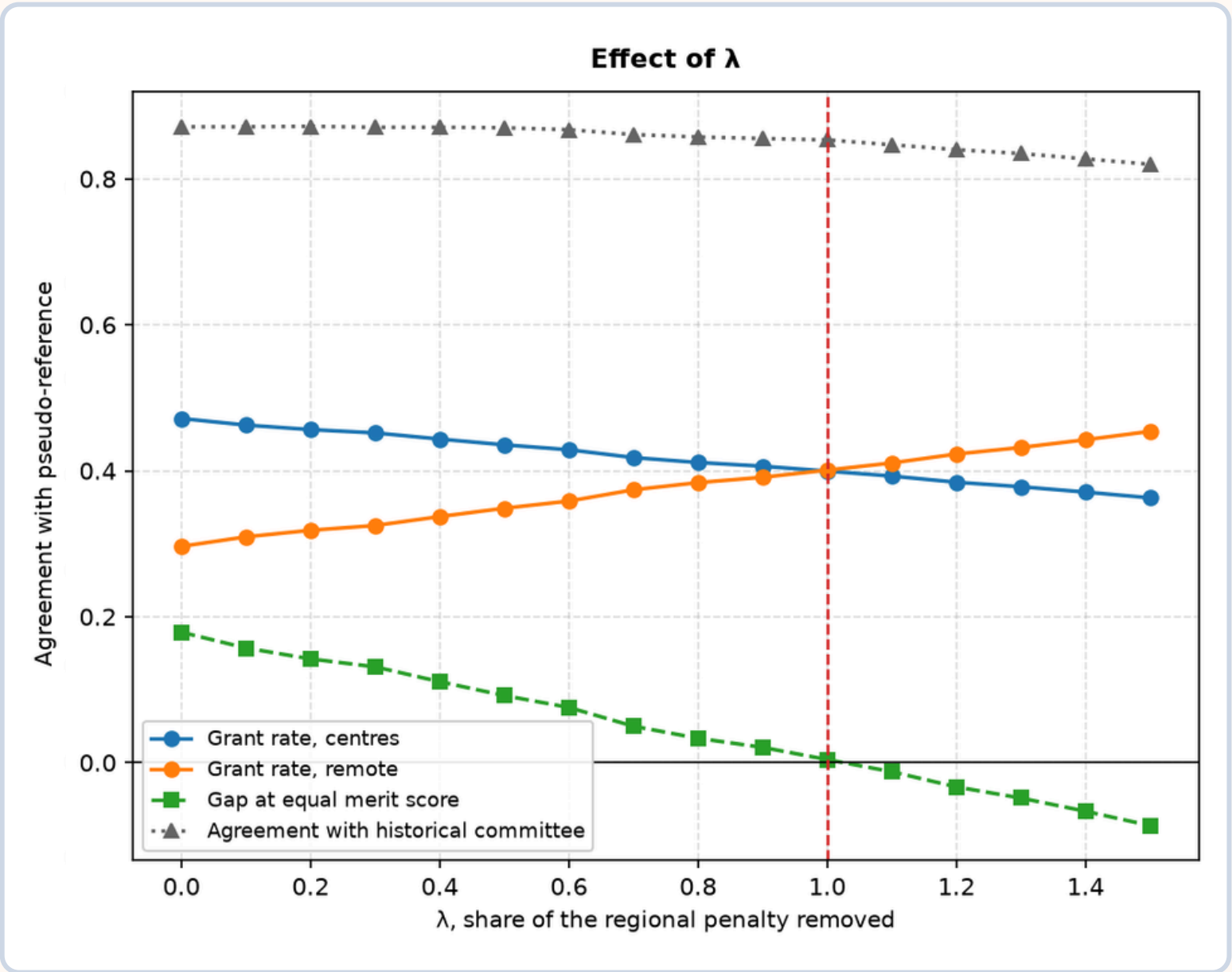
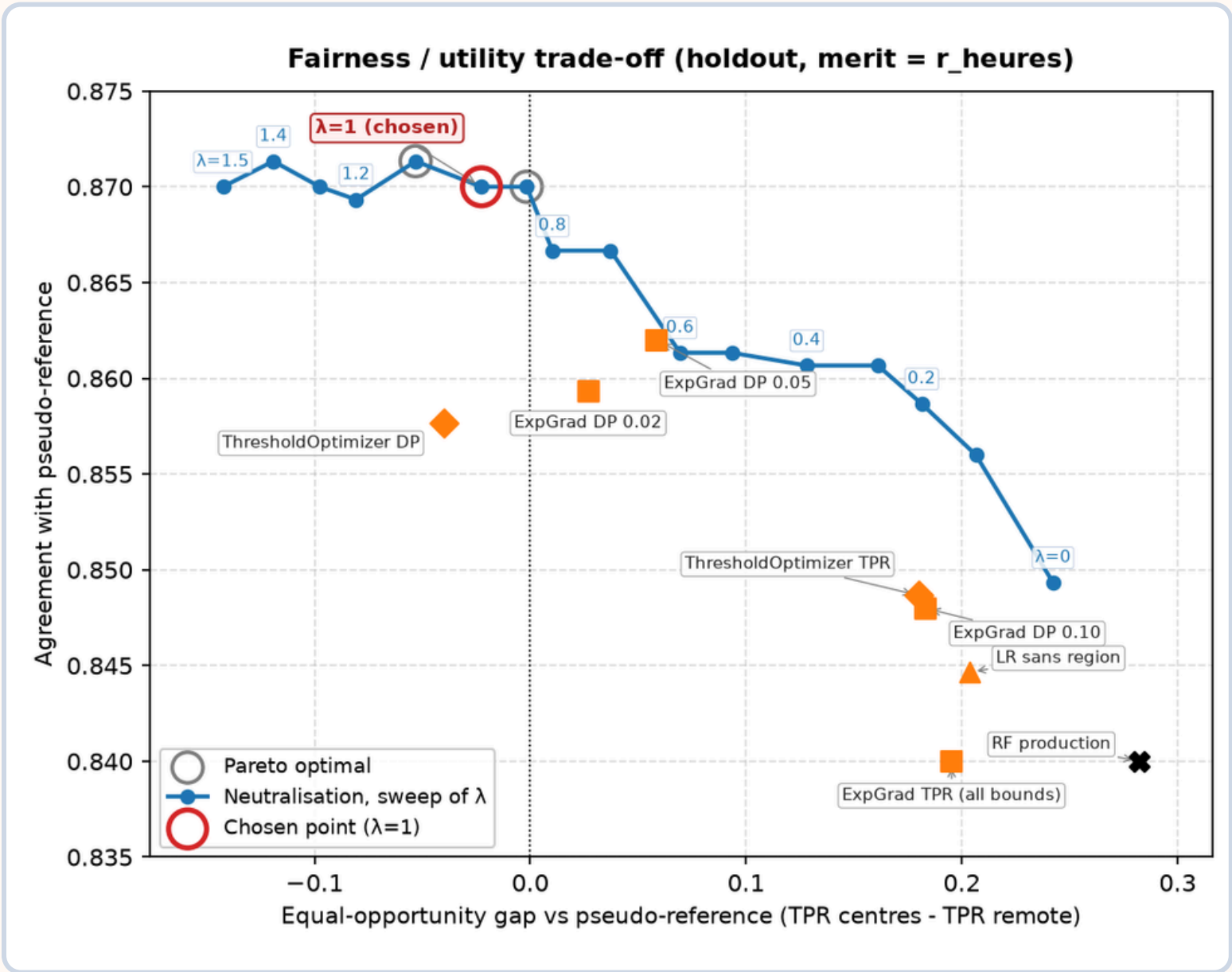
KEEP THE REGION TO REMOVE IT

- 1. Model the committee with an explicit remote indicator, so the penalty in one coefficient
- 2. Score with that penalty set to zero ($\lambda = 1$)
- 3. Grant the top 40% (the budget)

GRANT IF

$$R\text{ SCORE} + 0.145 \times \text{WEEKLY HOURS} \geq 30.08$$

Region, postal code, distance and income play no role in the decision.



FAIR AND MORE ACCURATE

Before (committee / production) → Ours

48% / 27% → 40.0% / 40.0%
GRANT RATE, CENTRES / REMOTE

0.27 → ≈ -0.02 (estimate)
EQUAL-OPPORTUNITY GAP

0.18 → 0.004
GRANT-RATE GAP BETWEEN REGIONS AT THE
SAME MERIT

88.7% → 94.6%
AGREEMENT WITH REFERENCE

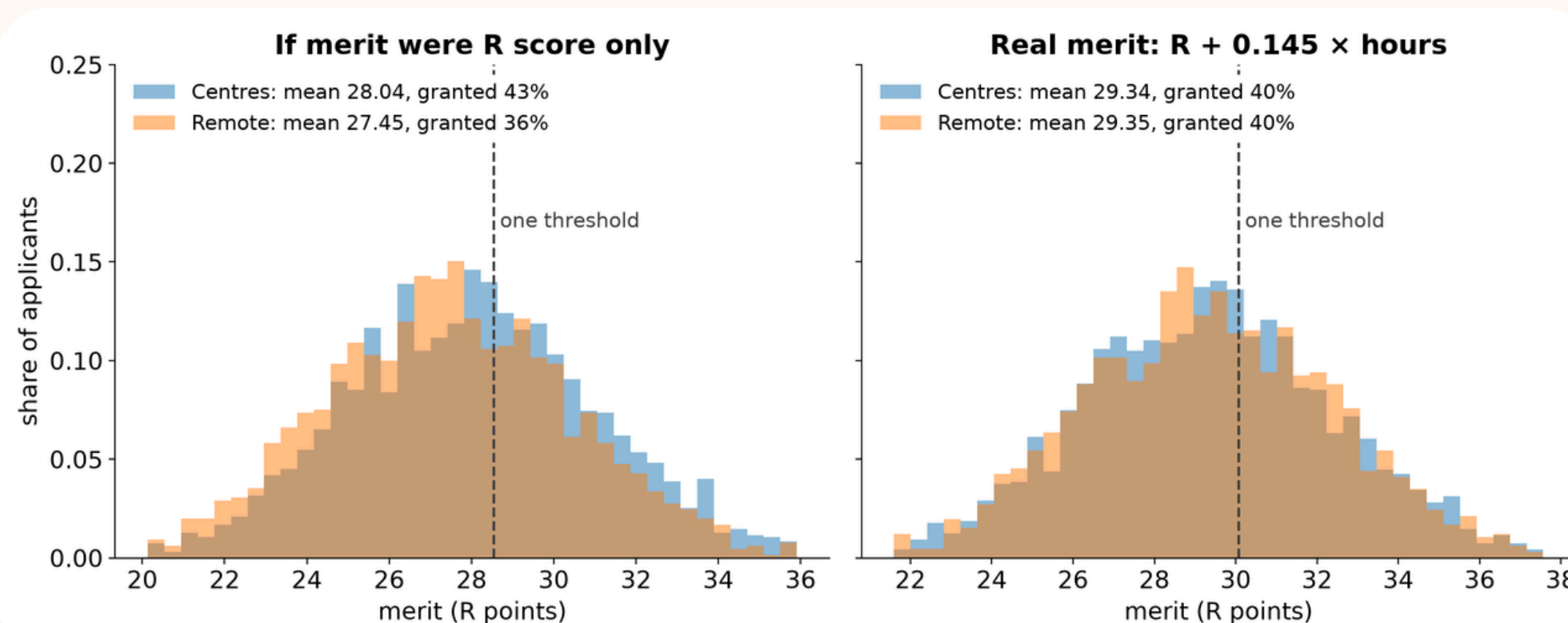
**EQUAL GRANT RATES (40.0% / 40.0%) COME FROM MEASURING MERIT CORRECTLY, NOT FROM A QUOTA.
THE NEXT SLIDE EXPLAINS WHY.**

WHY EQUAL OPPORTUNITY, NOT DEMOGRAPHIC PARITY?

Demographic parity: every region gets the same grant rate.

Equal Opportunity: a deserving applicant has the same chance, wherever they live.

1. A scholarship rewards merit. Equal Opportunity protects exactly that.
2. Demographic parity can force a different bar for each region. If merit were R alone, 40/40 would need $R \geq 28.79$ in the centres but only $R \geq 28.15$ in remote regions.
3. Demographic parity can be met without fixing anything: a 40/40 quota on the committee's own score keeps the income bias.



THIS WAS DISCRIMINATION, AND SOME CHOICES AREN'T OURS TO MAKE

- Quebec Charter, s. 10: the income bias falls under “social condition”; the regional bias acts as a proxy for it
- Four-fifths benchmark: committee 0.54 (region), 0.52 (income); ours 0.92 and 0.93
- The hours term helps remote, low-income and first-generation applicants; risk for students who cannot work
- The board decides, not the data team: compensating remote regions, the hours weight, separating scholarships from loans

~560

remote applicants refused in the past although they clear the corrected threshold (vs ~194 in the centres)

Four-fifths benchmark

Committee
0.54 (region)
0.52 (income)

Ours
0.92 (region)
0.93 (income)

BUILT TO BE AUDITED

- 2 inputs only (R score, hours): data minimisation under Law 25
- Exact explanation for every applicant (Law 25, s. 12.1): “you needed 0.85 more R points, or 5.9 more weekly hours”
- Human review band ± 0.25 points ($\approx 6\%$ of files), reviewers blind to region
- A label-free dashboard runs before every release; a red check blocks it, in code

Pre-release dashboard on five test scenarios					
Scenario	Historical committee	Our rule	Biased human reviewers	Population drift	Wrong model shipped
Result	HOLD	RELEASE	HOLD	HOLD	HOLD
			caught by the override check	told apart from bias	

ONE RULE. TWO INPUTS. ZERO REGIONAL PENALTY.

Audit · Fix · Monitor, all reproducible in `audit_rapport.ipynb` and `model_corrige.py`



[nghiep.qt/defi-equalgo](https://github.com/nghiepqt/defi-equalgo) · VNPHOEVER

APPENDIX

1. What if merit differs between regions in the future?
2. Why not a more complex model?
3. Isn't hours worked a proxy too?
4. Limitations

WHAT IF MERIT DIFFERS BETWEEN REGIONS IN THE FUTURE?

1

We keep one threshold for everyone (Equal Opportunity).

2

The dashboard's parity check flags the gap, and the board decides whether to compensate, through an affirmative action program under Quebec Charter s. 86.

WHY NOT A MORE COMPLEX MODEL?

1

On committee labels, a fully flexible model fits no better than the linear one ($p = 0.76$).

2

Tree models let the regional penalty leak into distance.

ISN'T HOURS WORKED A PROXY TOO?

Yes (AUC 0.81), but it raises remote applicants' chances and disadvantages no observed group.

LIMITATIONS

1

The EO gap is an estimate.

2

The data are synthetic.

3

**The legal analysis is ours,
not a lawyer's.**